

David LeBauer Ph.D.

agriculture, ecology, biogeochemistry

Summary

Scientist with over ten years of experience developing methods to make the best use of available information and technology in pursuit of more productive and sustainable agriculture and more efficient science.

Experience

Owner and Principal Scientist

Mar 2024–present

The LeBauer Approach, LLC

- Modeling Lead and Project Manager, California Cropland Measurement and Modeling Framework (Sept 2024–present).
- Scientific Advisor, Varaha Climate AG (Jun 2024–present).
- Scientific consultant.

Staff Scientist and Technical Lead, Model Validation and Calibration

Feb 2023–Feb

2024

Indigo Ag

- Technical lead for team that calibrated, validated, and wrote report following Climate Action Reserve Soil Enrichment Protocol to support generation of verified carbon credits.
- Wrote science requirements for automated model calibration and validation.
- Harmonized calibration and validation datasets of observed soil carbon, greenhouse gas fluxes, and yields; migrated dataset to a relational geospatial database.

Director of Data Science

Jul 2018–Jan 2023

University of Arizona, Division of Agriculture, Life and Veterinary Sciences, and Cooperative Extension

- Founded and built a data science support team comprising 4–6 data scientists and software engineers to provide research support for the Division of Agriculture, Life Science, and Cooperative Extension.
- Facilitated cross-disciplinary, collaborative research projects through data curation, statistical guidance, grant writing, workshops, and translation of research priorities across domains.
- Developed and launched a data science incubator program with outcomes that include preliminary results, upgraded scientific software, published papers, data, and code, and data visualization dashboards and analysis pipelines.
- Developed long-term vision and five-year plan using core USDA Hatch funding alongside over \$2m in extramural funding to support modeling, computing, and informatics projects from ARPA-E, DARPA, USDA, and NSF.

Senior Scientific Advisor

Jun 2015–Jul 2018

Agrible, Inc. (part-time)

- Designed and implemented analyses of yield data used to inform seed choice.

Contact

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Focus

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Honors & Awards

- **2023–2023:** Equity, Inclusion, and Diversity Award
North American Plant Phenotyping Network
- **2021–2021:** Best Paper
7th workshop on Computer Vision in Plant Phenotyping and Agriculture. ICCV 2021
- **2017–2017:** Outstanding Students Pushing Innovation (SPIN) Mentor
National Center for Supercomputing Applications
- **2014–2015:** NCSA Fellowship
National Center for Supercomputing Applications (\$25k)
- **2005–2007:** Graduate Fellowship
Kearney Soil Science Foundation (\$34k)
- **1998–1998:** Honors in Ecology and Graduation with Distinction
Duke University Department of Biology
- **1998–1998:** Benenson Award in the Arts
Duke University (\$1.5k)

Key Capabilities

- **PEcAn:** R package development; Bayesian statistics; ecosystem modeling; high-performance computing; mentoring
- **BETYdb:** SQL; PostGIS; UI design; APIs; networking; data management
- **TERRA REF:** Project management; cloud computing; API development; data standards; Python; MongoDB; netCDF; high-performance computing

- Consulted with software engineers and scientists to develop, analyze, and apply a crop model to inform and optimize farms and supply chains.
- Taught quantification and propagation of uncertainty in crop models, including soil physical properties, and advised on implementation of these methods.
- **BioCro**: R; GitHub; ecophysiology; project management; software architecture; netCDF; high-performance computing

Research Scientist, Carl R. Woese Institute for Genomic Biology

Aug 2012–Jun 2018

University of Illinois at Urbana-Champaign

- Data and Computing Pipeline Lead, TERRA-REF (2015–2019).
- Managed Energy Biosciences Institute modeling program (2012–2015) with \$600k annual budget and multi-institution team.
- Created and distributed PEcAn platform and database for ecophysiological and yield data synthesis; postdoctoral researcher (2009–2012).

Selected Data & Software

PEcAn Creator, developer, and co-PI (2009–present)

Ecoinformatics toolbox for model–data synthesis, analysis, and prediction integrating observations with understanding of crop and ecosystem functioning as an iterative learning process.

Skills: R package development; Bayesian statistics; ecosystem modeling; high-performance computing; mentoring

www.pecanproject.org github.com/PecanProject/pecan

BETYdb Creator and development lead (2009–present)

Database network and web interface to harmonize heterogeneous plant, ecosystem, and agronomic data for meta-analysis, simulation modeling, and synchronization among research teams.

Skills: SQL; PostGIS; UI design; APIs; networking; data management

www.betydb.org github.com/PecanProject/bety

TERRA REF Lead, data and computing pipeline (2015–2019)

Computing pipeline, reference data products, and cloud environments advancing high-throughput phenotyping for crop improvement.

Skills: Project management; cloud computing; API development; data standards; Python; MongoDB; netCDF; high-performance computing

www.terraref.org github.com/terraref

BioCro Development lead (2013–2015)

Dynamical simulation model combining physics, chemistry, and physiological processes to predict crop growth and water use.

Skills: R; GitHub; ecophysiology; project management; software architecture; netCDF; high-performance computing

github.com/ebimodeling/biocro

GHGVC Development lead (2013–2017)

Web interface and R package to compute biophysical and biogeochemical impacts of land use change.

github.com/ebimodeling/ghgvc

Education

PhD Earth System Science, University of California at Irvine 2008

Thesis: 'Carbon Balance of Ecosystems Under Nitrogen Enrichment'

MS Ecology, Agriculture concentration, University of California at Davis 2003

BS Biology, Duke University 1998

Selected Publications

- **2020** What does TERRA-REF's high resolution, multi sensor plant sensing public domain data offer the Computer Vision Community? Proc. IEEE/CVF International Conference on Computer Vision. doi: 10.1109/iccwv54120.2021.00162 — repository.arizona.edu/handle/10150/663361
- **2017** BETYdb: a yield, trait, and ecosystem service database applied to second-generation bioenergy feedstock production. Global Change Biology - Bioenergy. doi: 10.1111/gcbb.12420 — doi.org/10.1111/gcbb.12420
- **2013** Facilitating feedbacks between field measurements and ecosystem models. Ecological Monographs. doi: 10.1890/12-0137.1 — doi.org/10.1890/12-0137.1
- **2008** Nitrogen limitation of net primary productivity is globally distributed. Ecology. doi: 10.1890/06-2057.1 — doi.org/10.1890/06-2057.1